

KAVYA KOLICHELAM

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PROFESSIONAL SUMMARY

MS Computer Science (Data Science) graduate with 2+ years of experience in data analysis, machine learning, and AI-driven solution development. Skilled in Python, SQL, and end-to-end ML workflows — from data cleaning and EDA through model building, evaluation, and deployment. Hands-on experience with deep learning, NLP, LLMs, RAG, and Generative AI. Proven ability to build scalable data pipelines and dashboards, communicate insights to both technical and non-technical stakeholders, and deliver impactful results across large-scale datasets (200K+ records).

TECHNICAL SKILLS

Languages & Querying: Python, SQL, Java, C, C++

Machine Learning: Regression, Classification, Ensemble Methods (XGBoost, Random Forest), Feature Engineering, Model Evaluation, Cross-Validation, Scikit-learn

Deep Learning & CV: CNN, MobileNetV2, U-Net, Transfer Learning, Multi-Task Learning, TensorFlow, PyTorch

NLP & Generative AI: TF-IDF, Sentiment Analysis, NLTK, spaCy, LLMs, RAG, Prompt Engineering, OpenAI APIs

Data Analysis & Statistics: EDA, Statistical Analysis, Data Mining, Predictive Analysis, Trend Analysis, A/B Testing Concepts, Hypothesis Testing

Data Processing: Pandas, NumPy, SciPy, Data Cleaning, Data Validation, Data Wrangling, ETL Concepts, Feature Engineering

Visualization & BI: Power BI, Tableau, Matplotlib, Seaborn, Excel (Pivot Tables, VLOOKUP, XLOOKUP, Power Query), Google Sheets

Tools & Platforms: Git, GitHub, Jupyter Notebook, VS Code, Google Colab, MS Office Suite

PROFESSIONAL EXPERIENCE

Teaching Assistant – Machine Learning | Lawrence Technological University, Southfield, MI *Aug 2025 – May 2026*

- Guided 40+ students through ML model development, data preprocessing, feature engineering, and predictive modeling in a fast-paced academic environment
- Validated analytical outputs and student-built data pipelines; identified processing errors and guided corrective actions to improve result accuracy
- Communicated complex ML, statistical, and data concepts clearly to students with varying technical backgrounds, reinforcing analytical best practices
- Assisted with technical documentation, reporting, and cross-team coordination to meet semester deadlines

Machine Learning Engineer | Start Tech Innovations, India

Jan 2023 – Dec 2023

- Developed and deployed regression and classification models using Python, Scikit-learn, and XGBoost for production business workflows across structured datasets
- Performed extensive EDA, feature engineering, and statistical analysis to identify trends, anomalies, and optimization opportunities
- Evaluated models using cross-validation, precision, recall, ROC-AUC, and RMSE; iterated systematically to improve production performance
- Designed and automated scalable ML data pipelines, reducing manual processing time and ensuring reliable, repeatable outputs
- Collaborated with cross-functional teams to translate analytical findings into actionable recommendations for business stakeholders

Data Analyst Intern | Tech Solutions Pvt Ltd, Hyderabad, India

May 2022 – Dec 2022

- Analyzed large datasets using SQL and Python (Pandas, NumPy) to extract insights, identify trends, and support data-driven decision-making across departments
- Performed statistical analysis, data mining, and predictive analysis to uncover optimization opportunities and surface data quality issues
- Built and maintained Power BI dashboards and automated reports to enable self-service analytics for non-technical stakeholders
- Cleaned, validated, and reprocessed Excel and CSV datasets; investigated discrepancies and implemented fixes improving downstream reporting accuracy

- Collaborated with cross-functional teams to deliver insights supporting operational efficiency and KPI performance tracking

PROJECT EXPERIENCE

Neuro-Symbolic Hybrid Classification System | *Python, Scikit-learn, TF-IDF, Pandas, Rule-Based Logic*

- Designed a hybrid AI system combining rule-based logic with ML classifiers to process, clean, and classify 200,000+ news articles end-to-end, achieving 82.85% classification accuracy
- Built modular, scalable preprocessing and TF-IDF feature engineering pipeline with reusable components; generated detailed analytical reports on model performance and classification trends
- Demonstrated end-to-end pipeline design, large-scale data handling, and interpretable ML output — applicable across analyst, scientist, and engineering roles

Mental Health Sentiment Analysis | *Python, NLP, Scikit-learn, Matplotlib, Seaborn*

- Designed and implemented an end-to-end NLP pipeline to clean, preprocess, and classify sentiment in 38,000+ social media posts, applying TF-IDF vectorization and multiple ML classifiers
- Performed in-depth statistical analysis and model evaluation to ensure reliable results across imbalanced classes; produced clear visualizations using Matplotlib and Seaborn
- Communicated sentiment trends and analytical findings to both technical and non-technical audiences through structured visual reports

Hierarchical Deep Learning Framework for Concrete Crack Detection | *TensorFlow, MobileNetV2, U-Net, Transfer Learning*

- Engineered a two-stage deep learning pipeline: MobileNetV2 binary classifier (~90% accuracy) pre-filtered 85% of non-crack images before passing positives to a U-Net segmentation model (Dice score: 0.75)
- Applied transfer learning using pre-trained MobileNetV2 ImageNet weights, fine-tuned for domain-specific defect detection, significantly reducing training time and improving generalization
- Conducted systematic benchmarking and validation studies to confirm pipeline performance, computational efficiency, and real-world generalizability

LLM-Based AI Assistant | *OpenAI APIs, RAG, Prompt Engineering, Python*

- Designed and developed a Retrieval-Augmented Generation (RAG) chatbot using OpenAI APIs, implementing structured document indexing, semantic retrieval, and context-aware prompt engineering
- Integrated dynamic retrieval pipeline to ground LLM responses in relevant source documents, improving factual accuracy and reducing hallucinations
- Applied advanced prompt engineering techniques including chain-of-thought prompting and few-shot examples to optimize response quality across diverse query types

EDUCATION

Master of Science – Computer Science (Data Science)

Lawrence Technological University, Southfield, MI

GPA: 3.8

May 2026

Bachelor of Technology – Electrical & Electronics Engineering

Bhoj Reddy Engineering College for Women, Hyderabad, India

2019 – 2023

CERTIFICATIONS

- Machine Learning – Coursera (Andrew Ng)
- Deep Learning Specialization – Coursera
- Generative AI Fundamentals – Coursera
- Data Science Tools – IBM
- Python for Data Science – IBM
- SQL for Data Science – Coursera
- Data Analysis with Python – IBM / Coursera
- Data Visualization with Python – IBM / Coursera
- Excel Basics for Data Analysis – IBM / Coursera
- Power BI for Beginners – Microsoft / Coursera